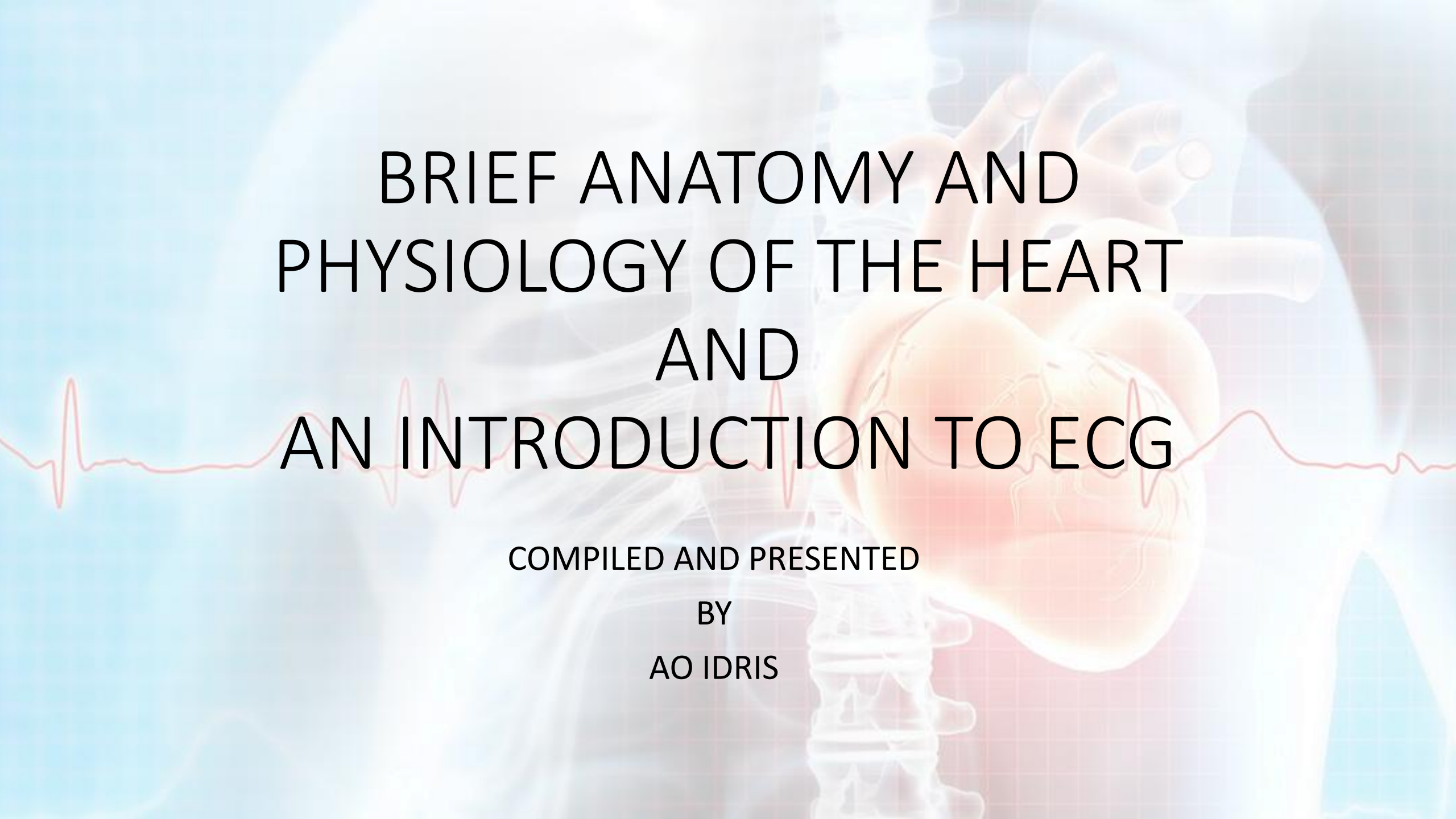


The background of the slide features a semi-transparent anatomical illustration of the human torso, focusing on the heart and lungs. A red ECG (heart rate) line is superimposed over the illustration, running horizontally across the middle of the slide. The text is centered over this graphic.

BRIEF ANATOMY AND PHYSIOLOGY OF THE HEART AND AN INTRODUCTION TO ECG

COMPILED AND PRESENTED

BY

AO IDRIS

Learning Objectives

- By the end of this session, participants should be able to:
- Describe the basic structure of the heart.
- Identify the main chambers, valves, and layers of the heart.
- Understand the cardiac conduction system and how it generates electrical impulses.
- Understand the basis of ECG.

Pre-presentation MCQs

1. What is the primary function of the heart?

- A) To filter blood
- B) To generate nerve impulses
- C) To pump blood throughout the body
- D) To produce red blood cells

2. Which of the following chambers pumps oxygenated blood into systemic circulation?

- A) Right atrium
- B) Right ventricle
- C) Left atrium
- D) Left ventricle

3. What is the name of the valve between the left atrium and left ventricle?

- A) Tricuspid valve
- B) Pulmonary valve
- C) Aortic valve
- D) Mitral (bicuspid) valve

4. Which of the following is NOT a typical indication for performing an ECG?

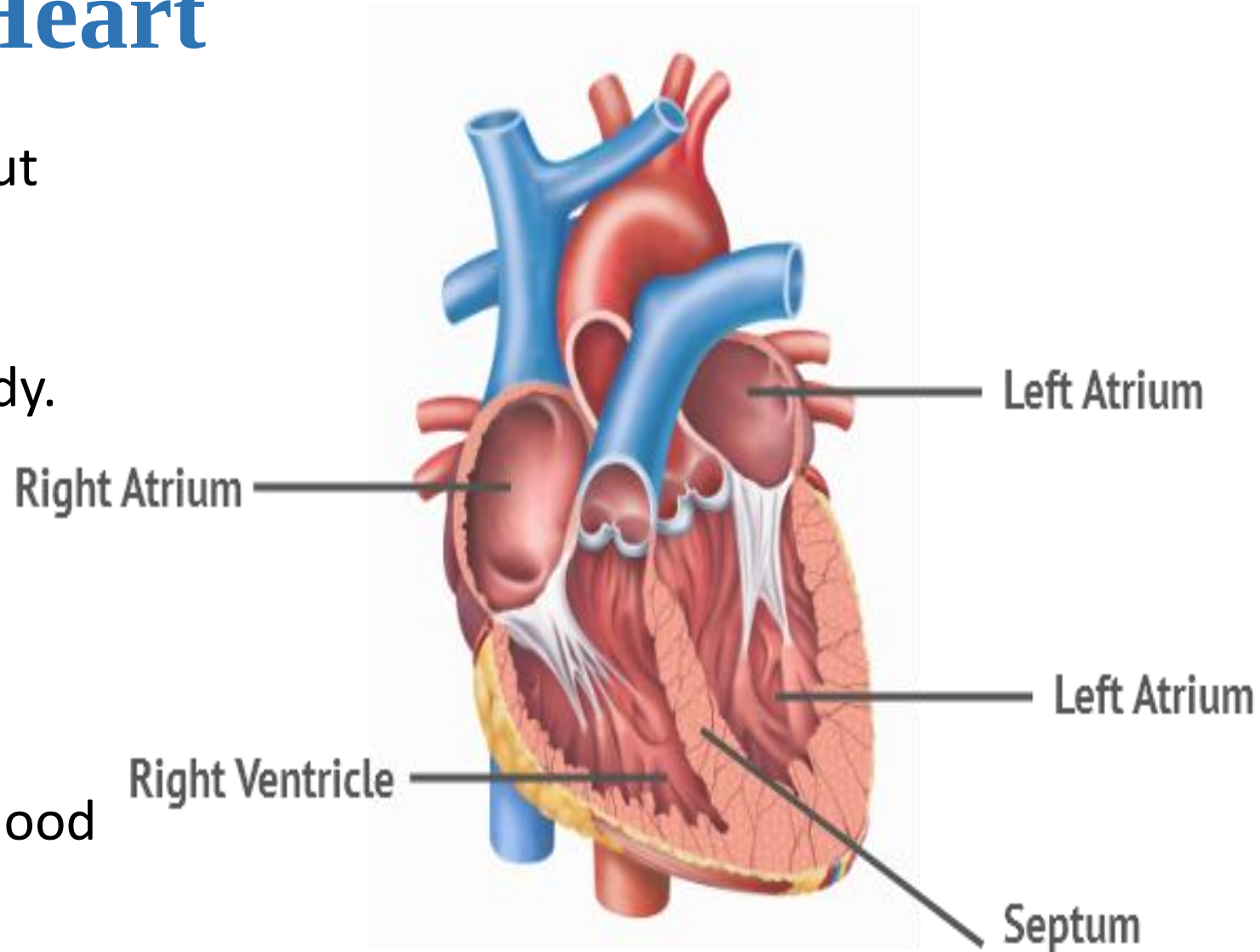
- A) Chest pain
- B) Syncope
- C) Sore throat
- D) Palpitations

5. What is an electrocardiograph?

- A) The process of performing an ECG
- B) The graphical output of heart activity
- C) The machine used to record the ECG
- D) The heartbeat itself

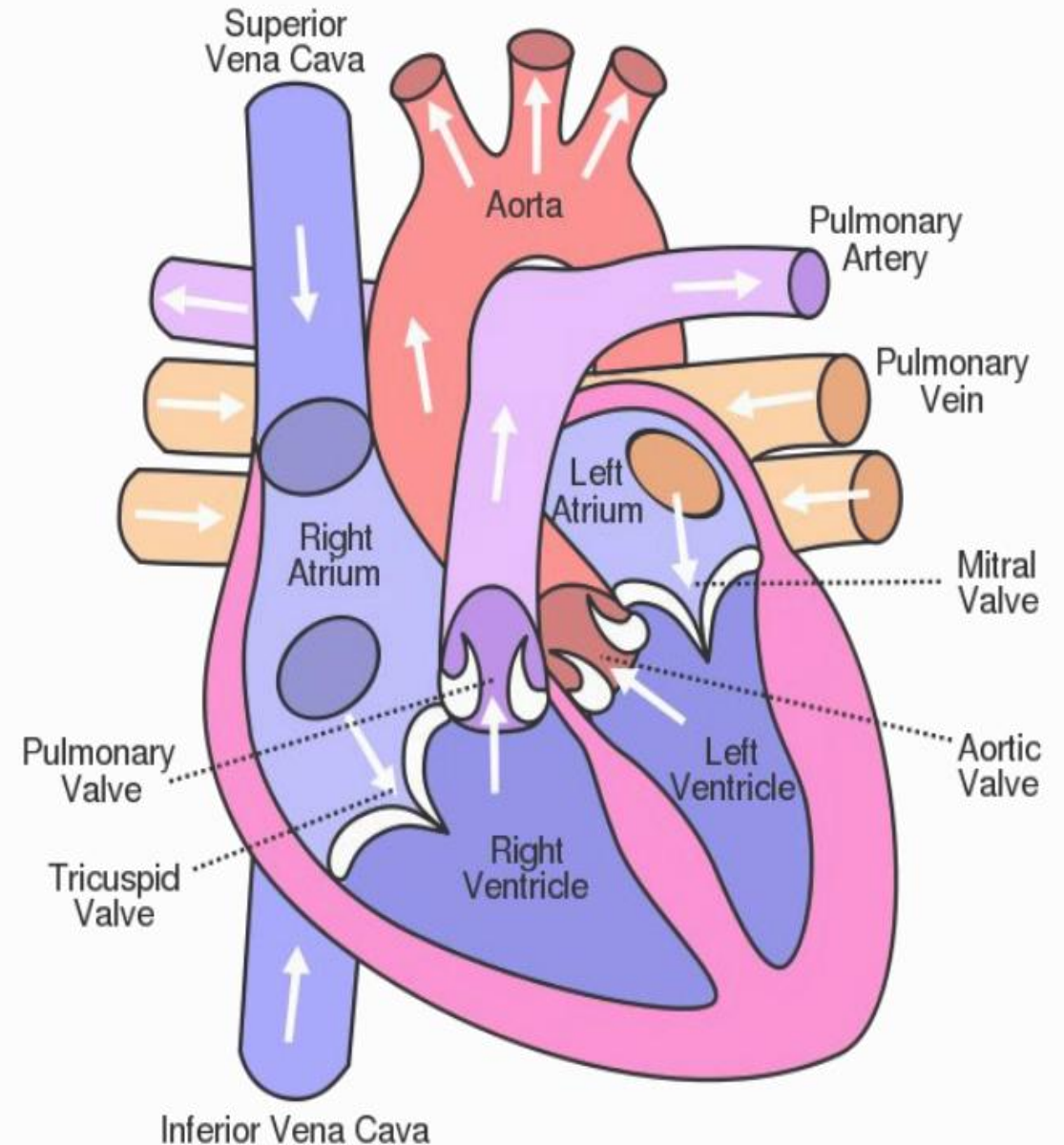
Overview of the Heart

- The heart is a muscular organ about the size of a fist, located in the thoracic cavity.
- It functions as a double pump to circulate blood throughout the body.
- Divided into four chambers:
 - Right Atrium (RA)
 - Right Ventricle (RV)
 - Left Atrium (LA)
 - Left Ventricle (LV)
- Separated by valves that control blood flow and prevent backflow.



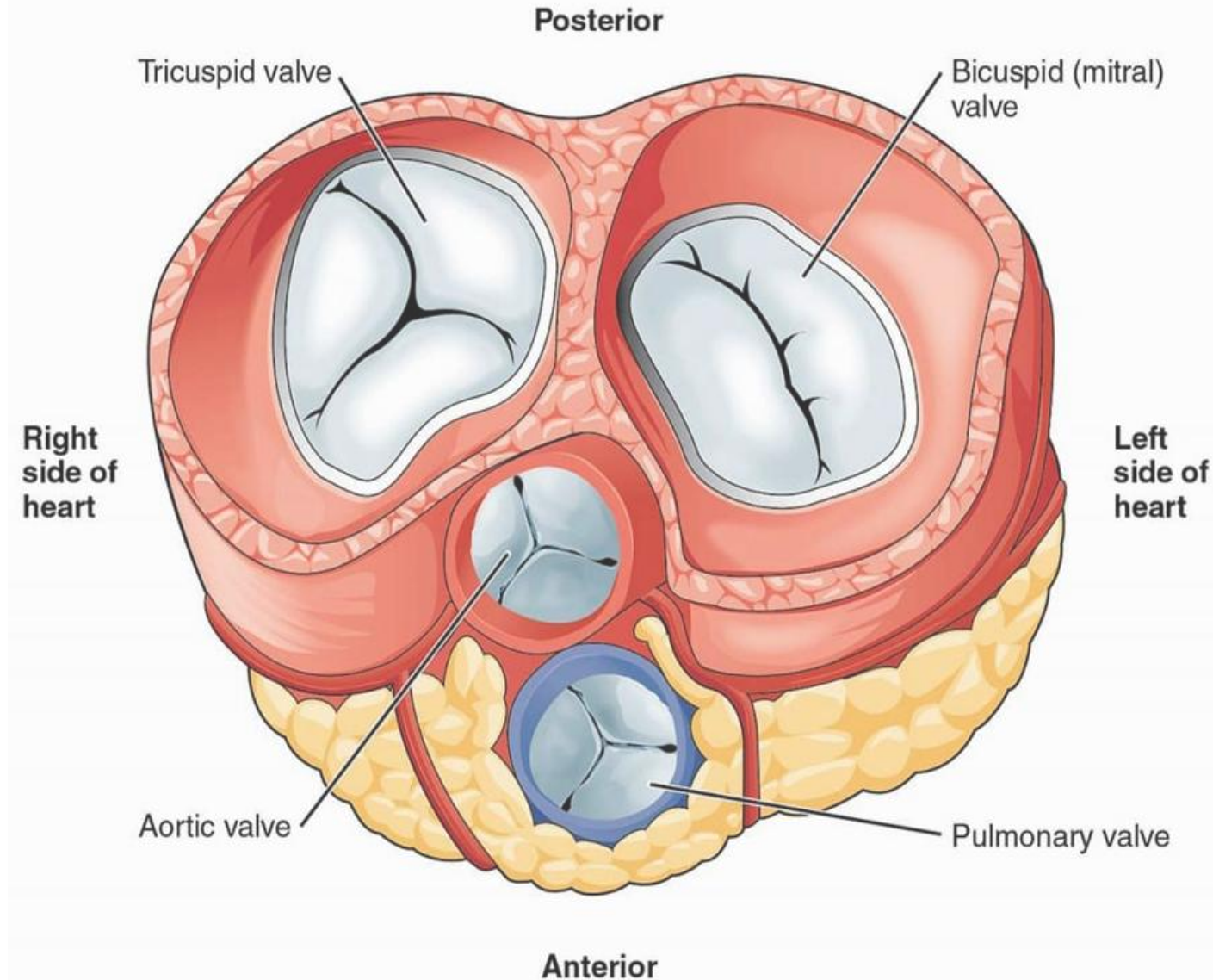
Circulation Pathways

- Right heart (pulmonary circulation):
- RA receives deoxygenated blood from the body → RV pumps it to the lungs via the pulmonary artery.
- Left heart (systemic circulation):
- LA receives oxygenated blood from the lungs → LV pumps it to the rest of the body via the aorta.
- This closed-loop system ensures tissues receive oxygenated blood and waste is removed.

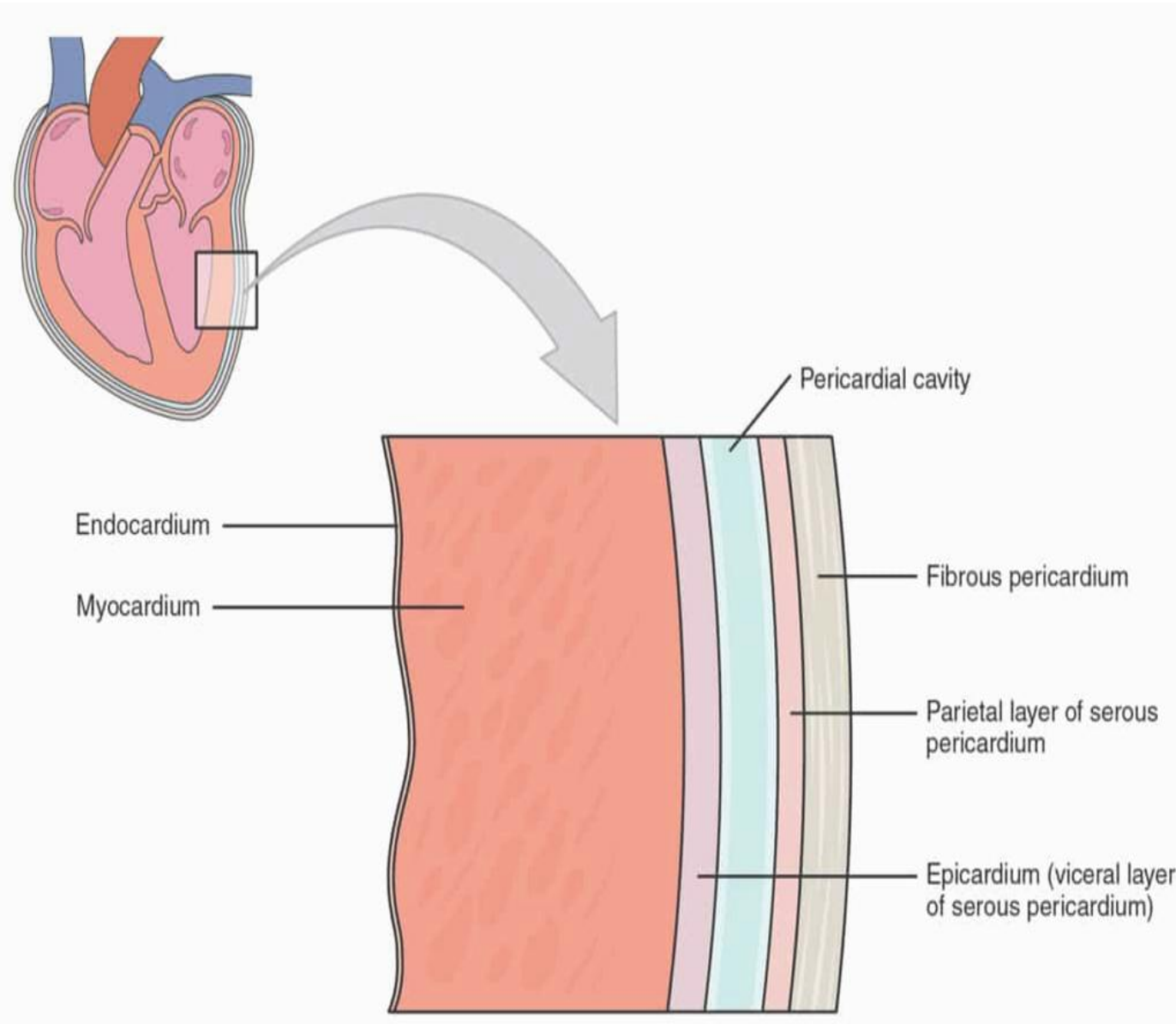


Heart Valves and Their Role

- Four valves ensure one-way flow:
- Tricuspid valve (RA → RV)
- Pulmonary valve (RV → pulmonary artery)
- Mitral valve (LA → LV)
- Aortic valve (LV → aorta)
- These open and close in response to pressure changes in the chambers.



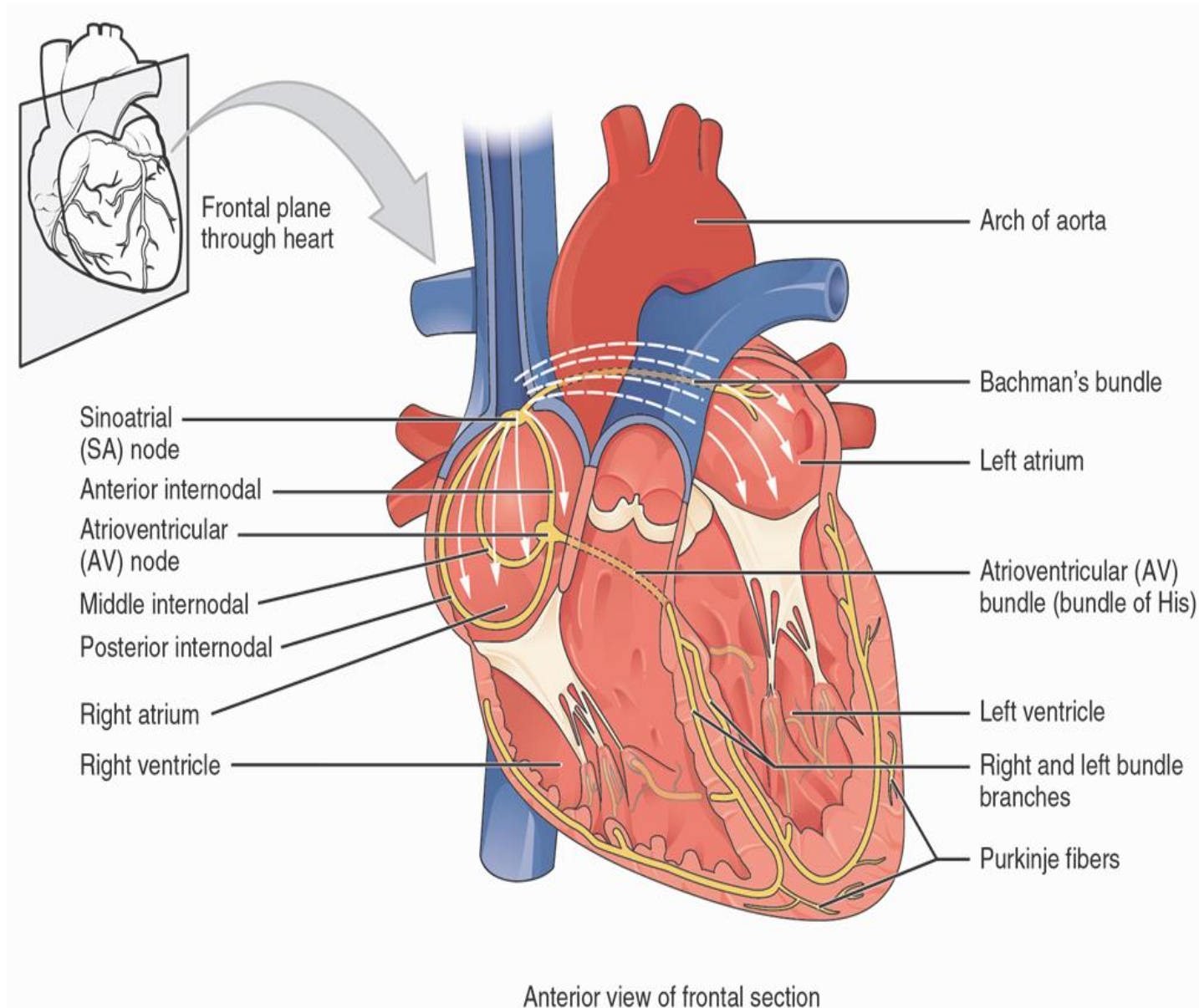
Layers of the Heart Wall



- Endocardium – Inner lining of the heart chambers.
- Myocardium – Thick muscular layer responsible for contraction.
- Epicardium – Outer surface of the heart.
- Pericardium – A protective double-layered sac encasing the heart.
- ECG measures electrical activity in the myocardium, so understanding this layer is essential.

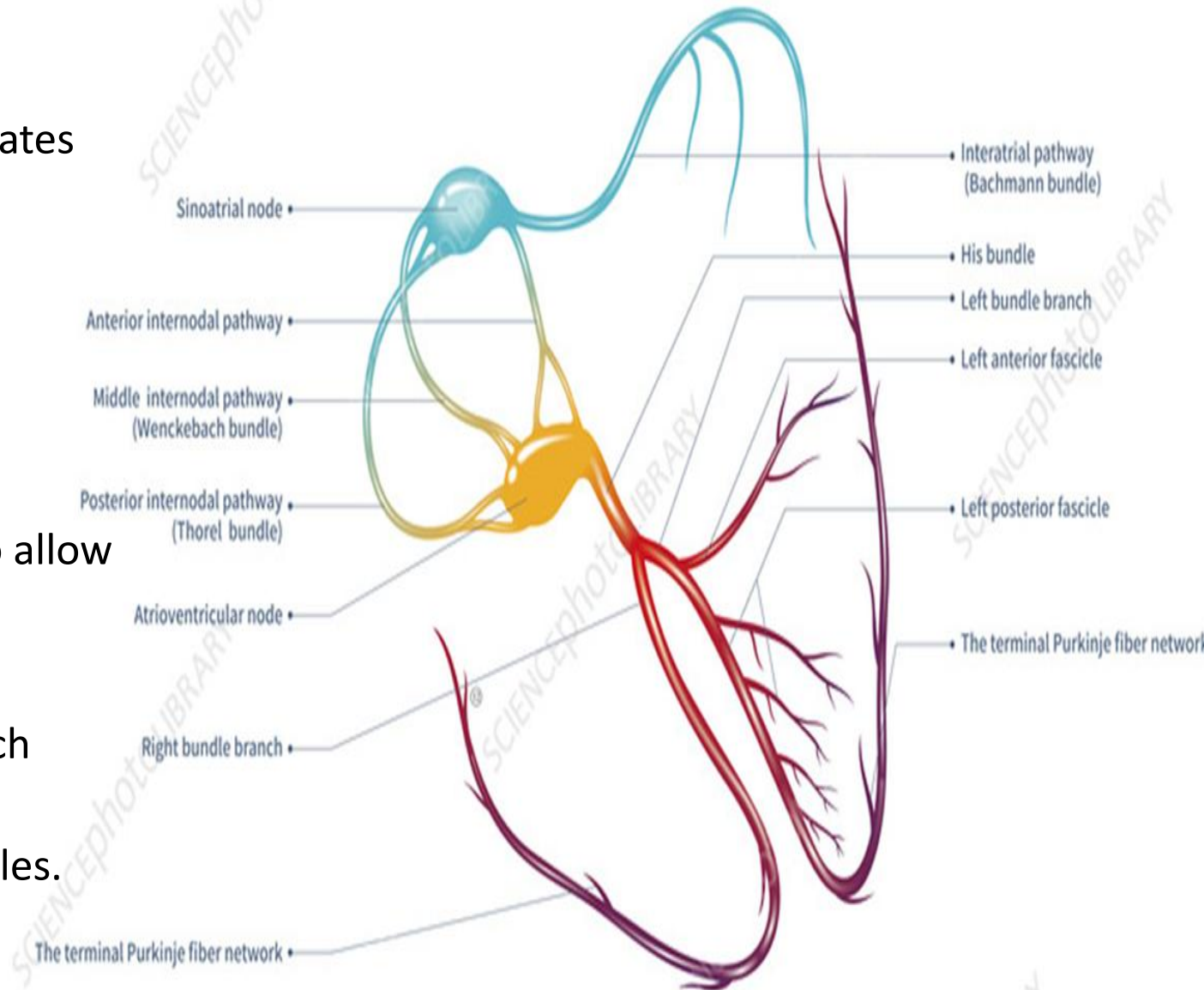
Function Histology of the Heart

- The cardiac tissue are divided into conductive tissue and contractile tissue in the myocardium.
- The conductive (nodal) tissue comprises special modified nerve cells that initiate and conduct rhythmic depolarisations of the myocardial cells.
- Nodal cells: SA Node, intermodal pathways, AV node, AV bundle (Bundle of His), Bundle branches (left and right).
- Contractile cells: Actin, Myosin, Tropomyosin, Sarcoplasmic reticulum



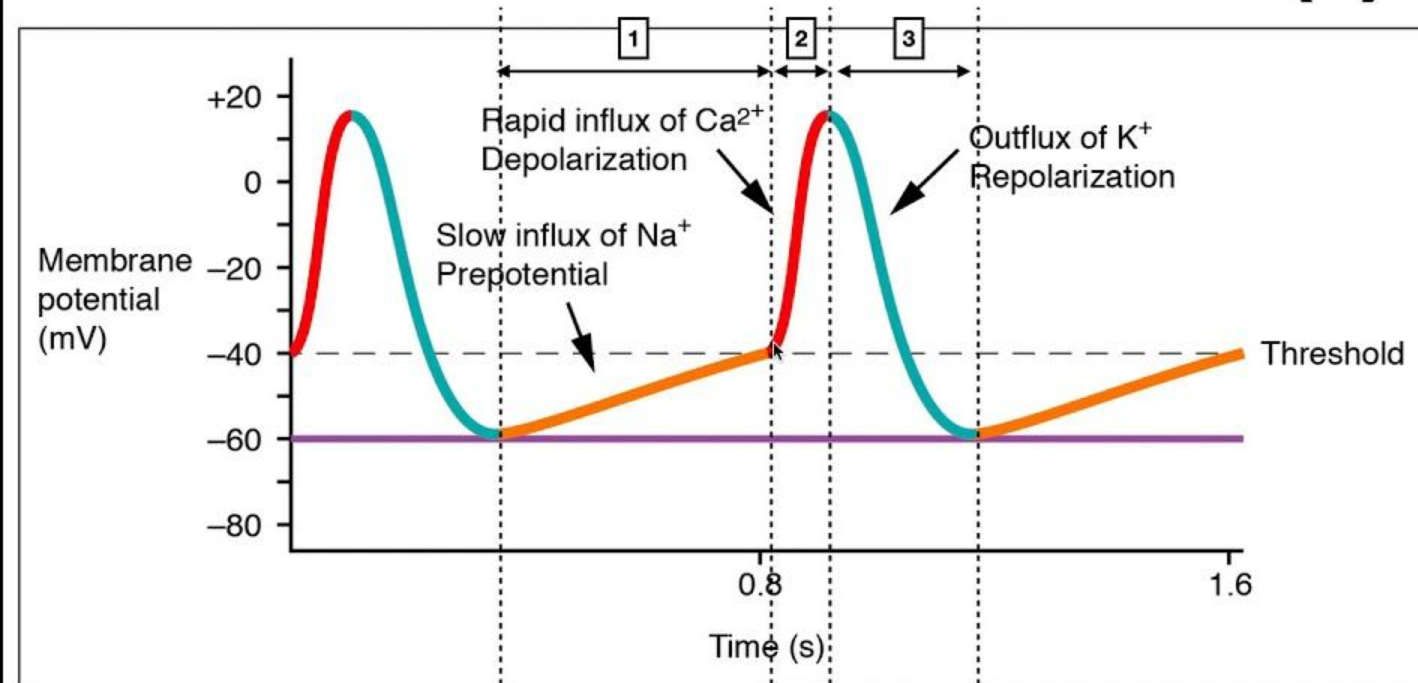
The Cardiac Conduction System

- The heart has an intrinsic electrical system to generate and propagate impulses.
- Key components:
- SA node (Sinoatrial node): natural pacemaker, initiates heartbeat.
- Internodal pathways:
 - i. Bachmanns bundle
 - ii. Wencheback
 - iii. Thorel
- AV node (Atrioventricular node): delays impulse to allow atrial contraction.
- Bundle of His: transmits impulse to ventricles.
- Right and Left Bundle Branches: conduct down each ventricle.
- Purkinje fibers: spread impulse throughout ventricles.

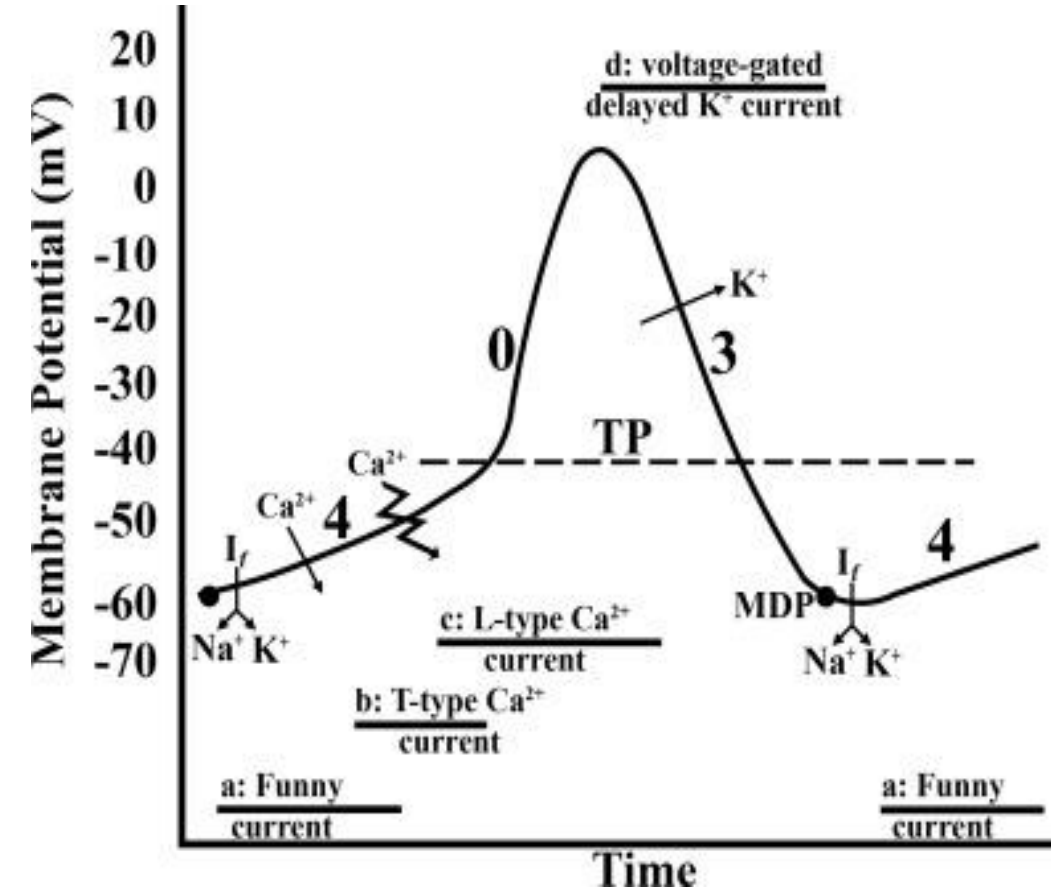


Action Potential in nodal cell

Cardiac Pacemaker Potential (II)

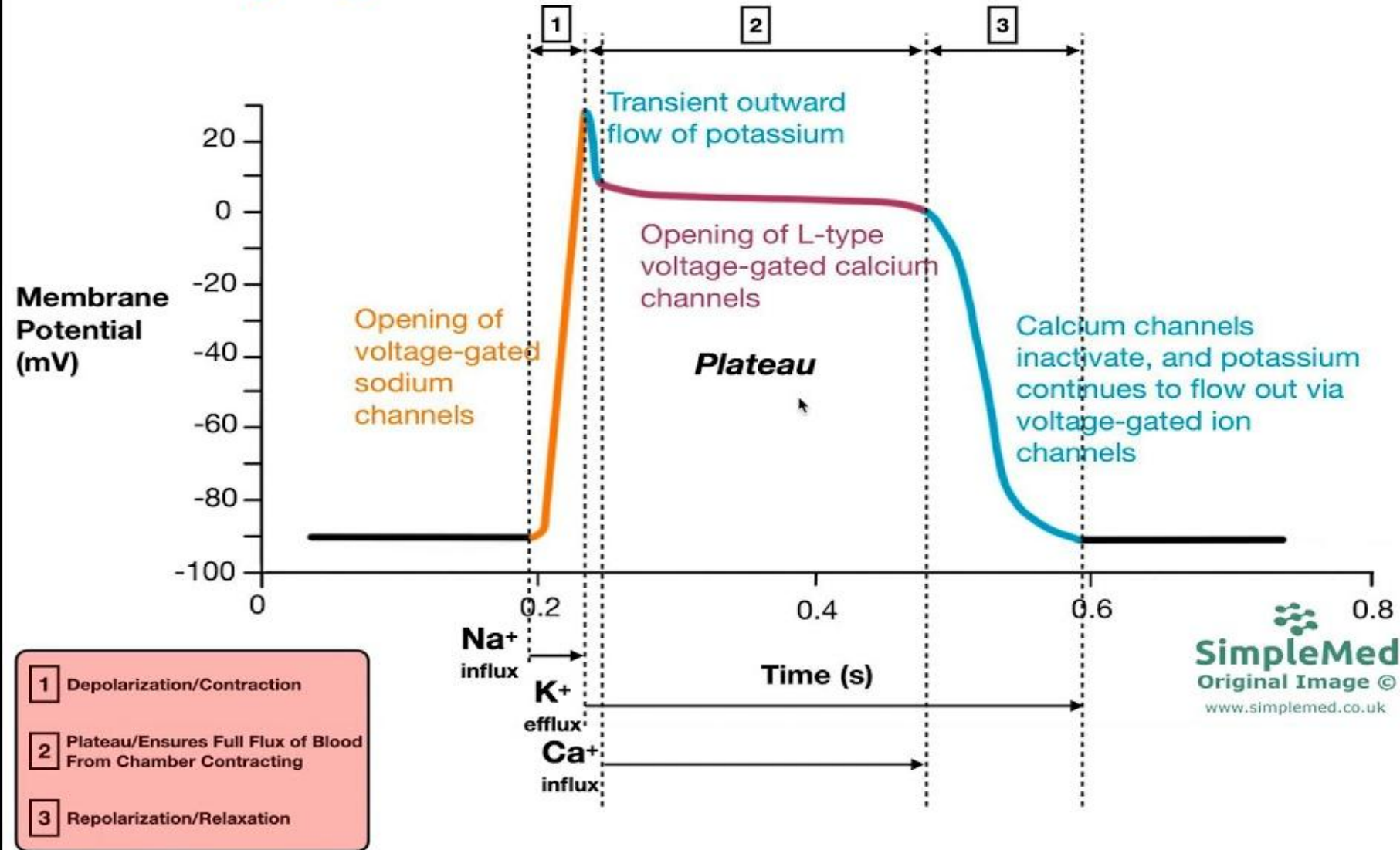


- Frequency of depolarizations [2] sets the heart rate.
- This is controlled by the heart's pacemaker, a cluster of cells in the right atrium called the **sinoatrial node** (SA node).
- Rate can be modulated by the **Vagus nerve** (↓HR) or **epinephrine** (↑HR).

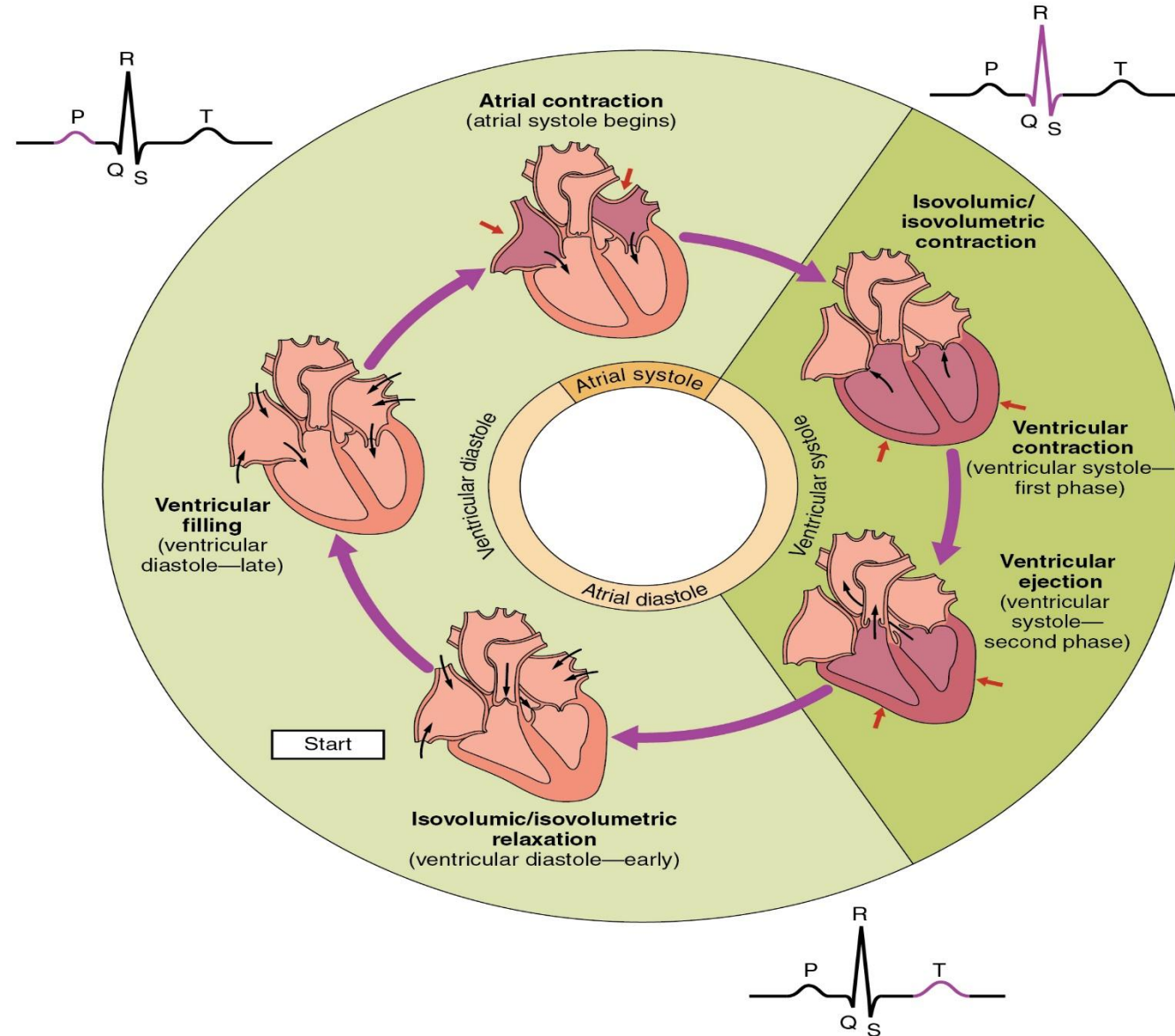


Action potential in contractile cell

Cardiomyocyte Action Potential: One Individual Cell

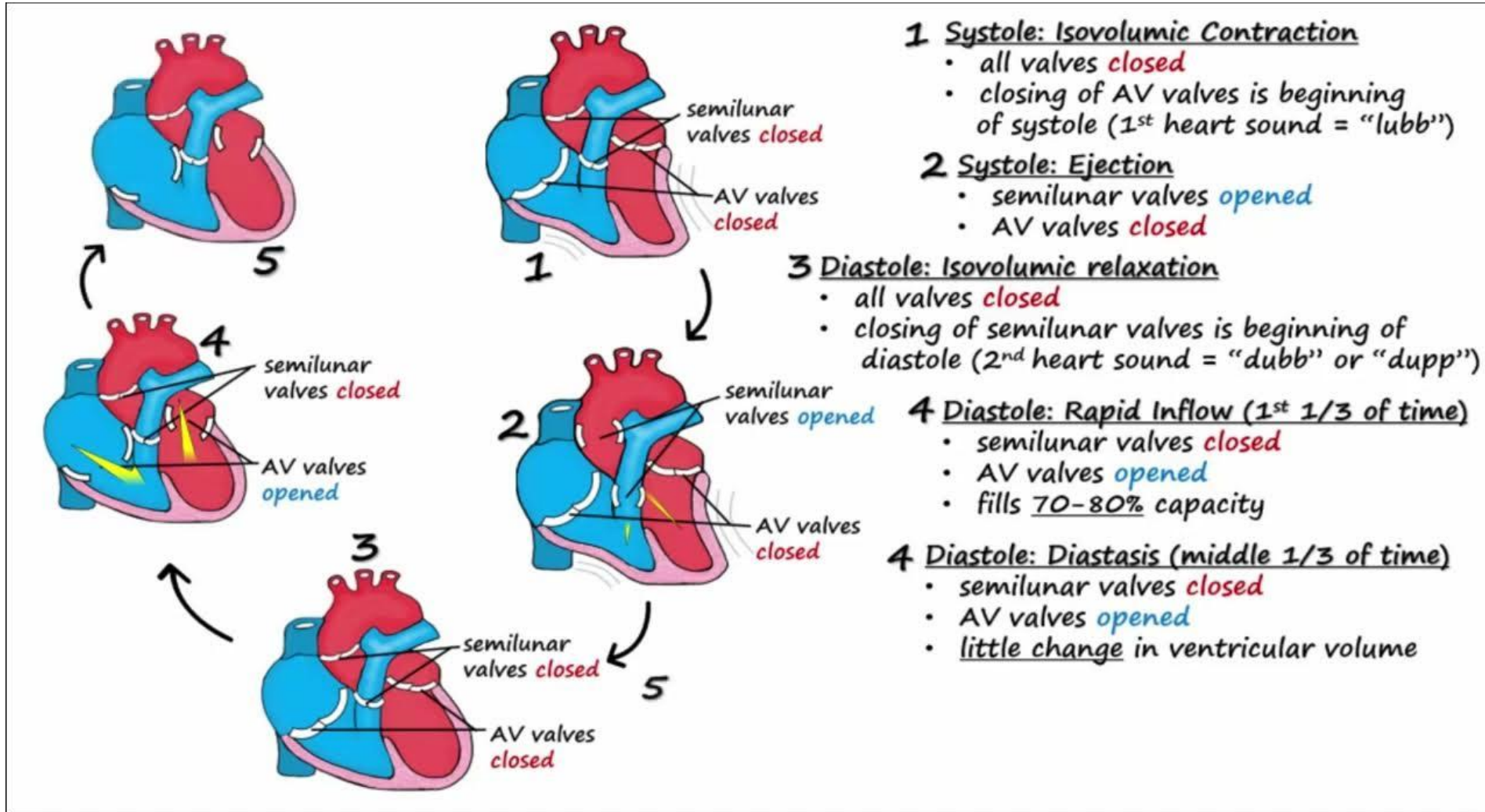


Electrical Activity and Cardiac Cycle

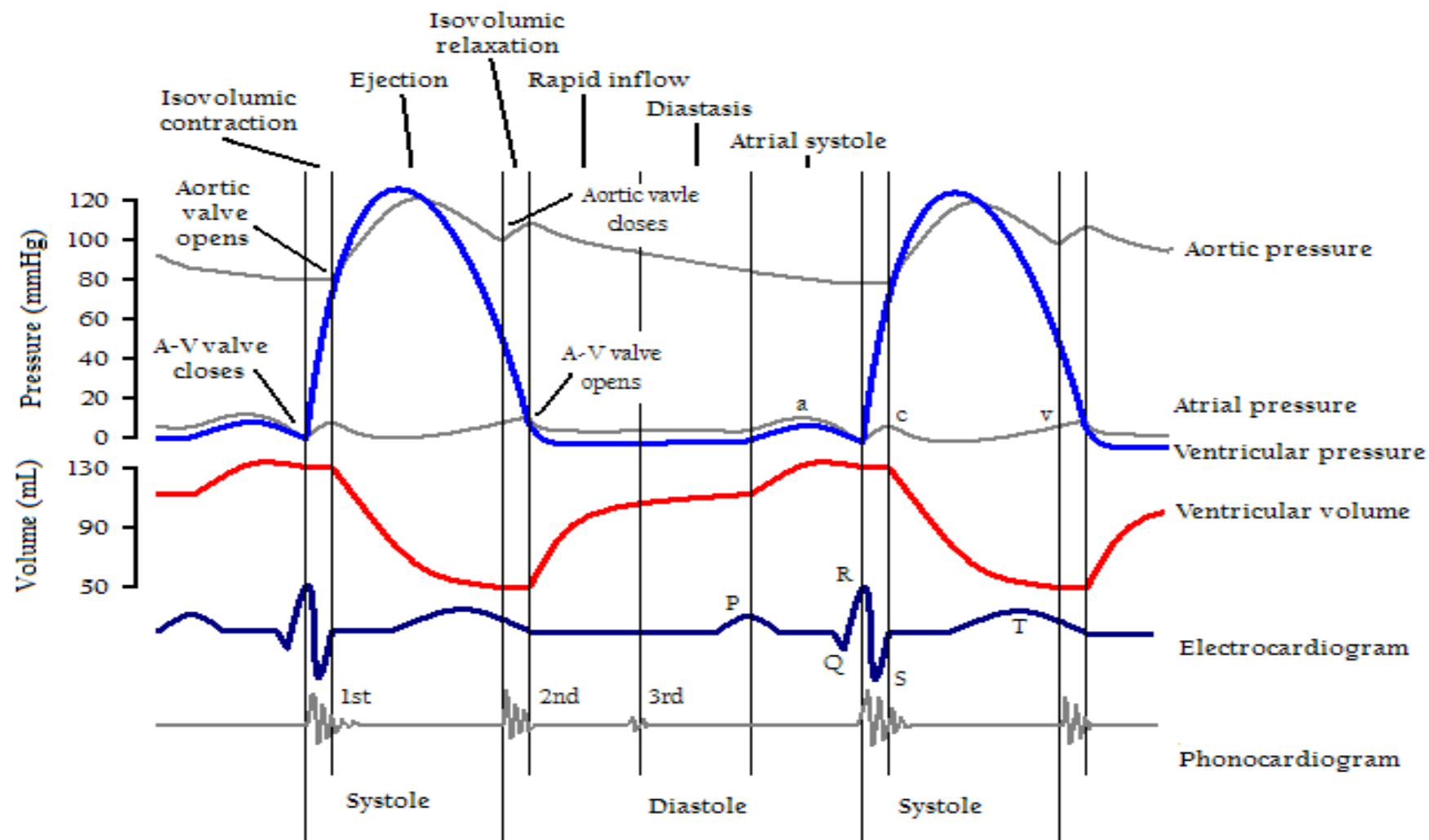


- The heart undergoes:
- Depolarization – electrical activation → muscle contraction.
- Repolarization – return to resting state → muscle relaxation.
- Systole = Contraction, Diastole = Relaxation
- These phases are reflected in the ECG waveform.

Cardiac Cycle



Cardiac Cycle



Introduction to ECG: Brief History

- **1849:** Emil du Bois-Reymond first discovers electrical activity in animal hearts.
- **1887:** Augustus Waller records the first human heart signal using a capillary electrometer.
- **1903:** Willem Einthoven invents the first practical **electrocardiograph** (uses the string galvanometer).
- He introduces the **P, Q, R, S, T** wave nomenclature still used today.
- Awarded **Nobel Prize** in **Physiology/Medicine** in 1924.
- Modern ECGs are digital, portable, and essential in diagnostics.



Introduction to ECG

- Electrocardiography: The process of recording the heart's electrical activity using an ECG machine.
- Electrocardiograph: The instrument/device used to record the ECG. It amplifies and displays electrical signals from the heart.
- Electrocardiogram: The actual recording or tracing produced by the electrocardiograph.

Indications for ECG

Diagnostic Purposes:

Chest pain (to rule out myocardial infarction)

- I. Palpitations or irregular heartbeat
- II. Syncope (fainting) or dizziness
- III. Shortness of breath or unexplained fatigue
- IV. Suspected electrolyte imbalances (e.g., hyperkalemia)
- V. Preoperative assessment in patients with cardiovascular risk.

• Monitoring Purposes:

- I. Ongoing cardiac conditions (e.g., arrhythmias, ischemia)
- II. Response to medications (e.g., antiarrhythmics, digoxin, QT-prolonging drugs)
- III. Pacemaker function
- IV. Post-MI or post-surgical follow-up

Contraindications and Limitations

There are no absolute contraindications to performing a standard 12-lead ECG.

However, caution is needed in the following cases:

- Skin infections or burns at electrode sites
- Severe trauma or fractures (e.g., rib fractures that make lead placement painful)

Limitations of ECG:

- May appear normal in early or small infarctions
- Not reliable for detecting mechanical problems (e.g., valve disease, heart failure) — needs echo
- Interference from movement, tremors, or loose electrodes can distort tracings
- Does not measure cardiac output or pump function
- Silent arrhythmias may be missed in a single snapshot

Common diagnosis made with ECG

1. Arrhythmias

- Atrial fibrillation
- Atrial flutter
- Ventricular tachycardia/fibrillation
- Bradycardia, Tachycardia, AV blocks

2. Myocardial Ischemia or Infarction

ST-elevation MI (STEMI)

Non-ST elevation MI (NSTEMI)

Old infarcts (Q waves)

3. Electrolyte

ImbalancesHyperkalemia (peaked T

waves)

Hypokalemia (U waves, flat T)

Calcium abnormalities (QT changes)

4. Conduction Abnormalities

Bundle branch blocks (LBBB/RBBB)

AV blocks (1st, 2nd, 3rd degree)

5. Drug Effects or Toxicity

Digoxin effect

QT prolongation from medications

Clinical Relevance

- Understanding cardiac anatomy and physiology is critical for:
- Correct ECG lead placement.
- Identifying abnormal rhythms (e.g., atrial fibrillation, heart block).
- Recognizing ischemia or infarction patterns (e.g., ST elevation).
- A solid foundation here aids accurate interpretation and diagnosis.

Summary

- The heart is a four-chambered pump with a unidirectional flow system.
- Electrical impulses arise from specialized cells and travel in a predictable pattern.
- The ECG records these electrical events, which reflect underlying anatomy and physiology.
- Basic knowledge of the heart's structure helps make sense of complex ECGs.

Q&A

Which part of the conduction system delays the impulse to allow ventricular filling?

- A) SA Node
- B) AV Node
- C) Purkinje Fibers
- D) Left Bundle Branch

References / Further Resources

- Books:
- "Rapid Interpretation of EKGs" by Dale Dubin
- "Anatomy & Physiology Made Incredibly Easy"
- Websites:
- www.ecglibrary.com
- www.teachmeanatomy.info
- <https://smart.servier.com>

